Report

Problem Statement

Develop a C++ program to represent webpages as vertices and hyperlinks as directed edges. Apply DFS from a selected webpage and display the visitation order. Display the pages reached from the starting page and identify pages not visited during the traversal.

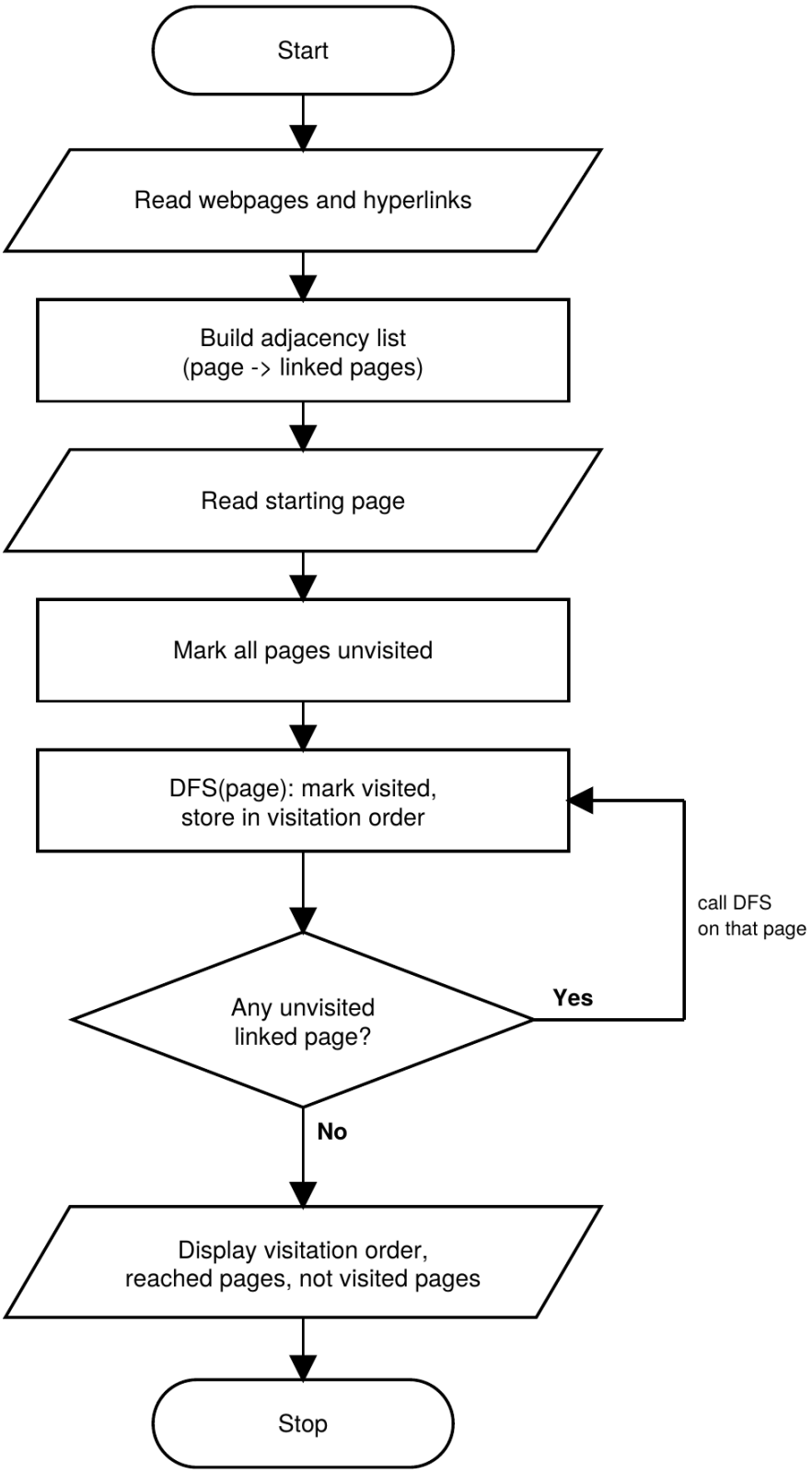
Objective

To implement a C++ program that models webpages as vertices and hyperlinks as directed edges, applies DFS from a chosen starting page, displays the visitation order, lists the reachable pages, and identifies the unvisited (unreachable) pages.

Algorithm

1. Start.
2. Read the number of webpages and their names.
3. Read the number of hyperlinks.
4. For each hyperlink, read (from, to) and add a directed edge from -> to in the adjacency list.
5. Display the webpages with their hyperlinks.
6. Read the starting page.
7. Mark all pages as unvisited.
8. Call DFS(start): mark the page visited and store it in the visitation order.
9. For each unvisited page linked from the current page, call DFS recursively.
10. Display the visitation order, the pages reached from the start, and the pages with visited = false as not visited.
11. Stop.

Flowchart


Start
Read webpages and hyperlinks
Build adjacency list
(page -> linked pages)
Read starting page
Mark all pages unvisited
DFS(page): mark visited,
store in visitation order
Any unvisited
linked page?
Yes
call DFS
on that page
No
Display visitation order,
reached pages, not visited pages
Stop